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Measuring the Cost of Living

Principles: Chapter 24

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The Consumer Price Index

- Consumer price index (CPI)
  - Measure of the cost of living
    - For a typical consumer
  - Reported monthly by the Bureau of Labor Statistics
    - Track price changes for on a fixed basket of goods and services

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Table 1

Calculating the Consumer Price Index and the Inflation Rate: An Example

This table shows how to calculate the consumer price index and the inflation rate for a hypothetical economy in which consumers buy only hot dogs and hamburgers.

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

Basket = 4 hot dogs, 2 hamburgers

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2010	\$1	\$2
2011	2	3
2012	3	4

Step 3: Compute the Cost of the Basket of Goods in Each Year

2010	(\$1 per hot dog × 4 hot dogs) + (\$2 per hamburger × 2 hamburgers) = \$8 per basket
2011	(\$2 per hot dog × 4 hot dogs) + (\$3 per hamburger × 2 hamburgers) = \$14 per basket
2012	(\$3 per hot dog × 4 hot dogs) + (\$4 per hamburger × 2 hamburgers) = \$20 per basket

Step 4: Choose One Year as a Base Year (2010) and Compute the Consumer Price Index in Each Year

2010	(\$8 / \$8) × 100 = 100
2011	(\$14 / \$8) × 100 = 175
2012	(\$20 / \$8) × 100 = 250

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year

2011	(175 − 100) / 100 × 100 = 75%
2012	(250 − 175) / 175 × 100 = 43%

Calculating CPI

Chose a base year and compute the CPI

- Base year = benchmark
  - Price of basket of goods & services in current year
  - Divided by price of basket in base year
  - times 100

Compute the inflation rate

$$\text{Inflation rate in year 2} = \frac{\text{CPI in year 2-CPI in year 1}}{\text{CPI in year 1}} \times 100$$

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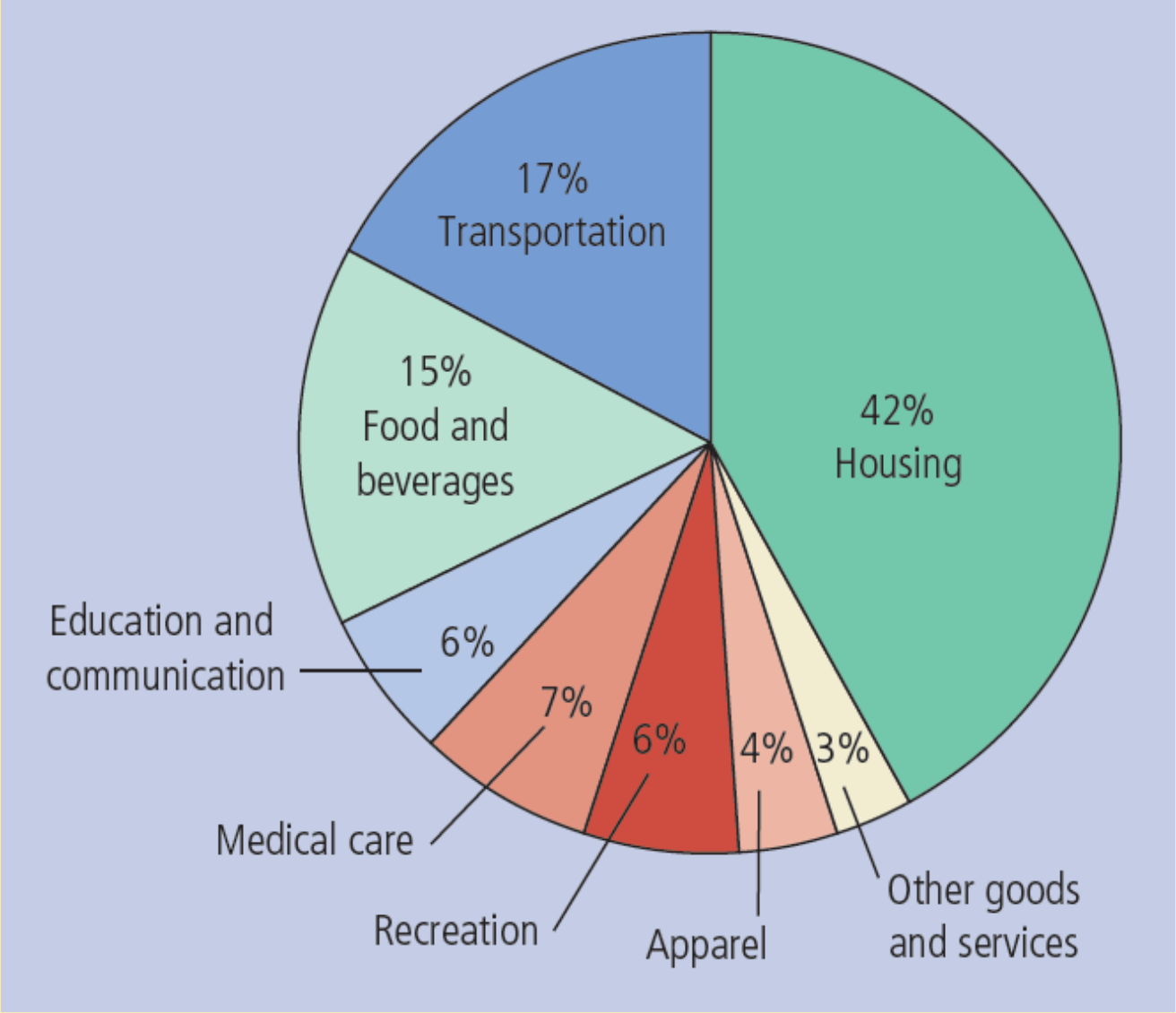
Figure 1

The Typical Basket of Goods and Services



This figure shows how the typical consumer divides spending among various categories of goods and services. The Bureau of Labor Statistics calls each percentage the "relative importance" of the category.

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The Consumer Price Index

- Problems in measuring the cost of living
  - Fixed basket, irres. of demand
    - Prices do not change proportionately
    - Consumers substitute toward goods that have become relatively less expensive
  - Introduction of new goods
    - More variety of goods
  - Unmeasured quality change
    - Changes in quality

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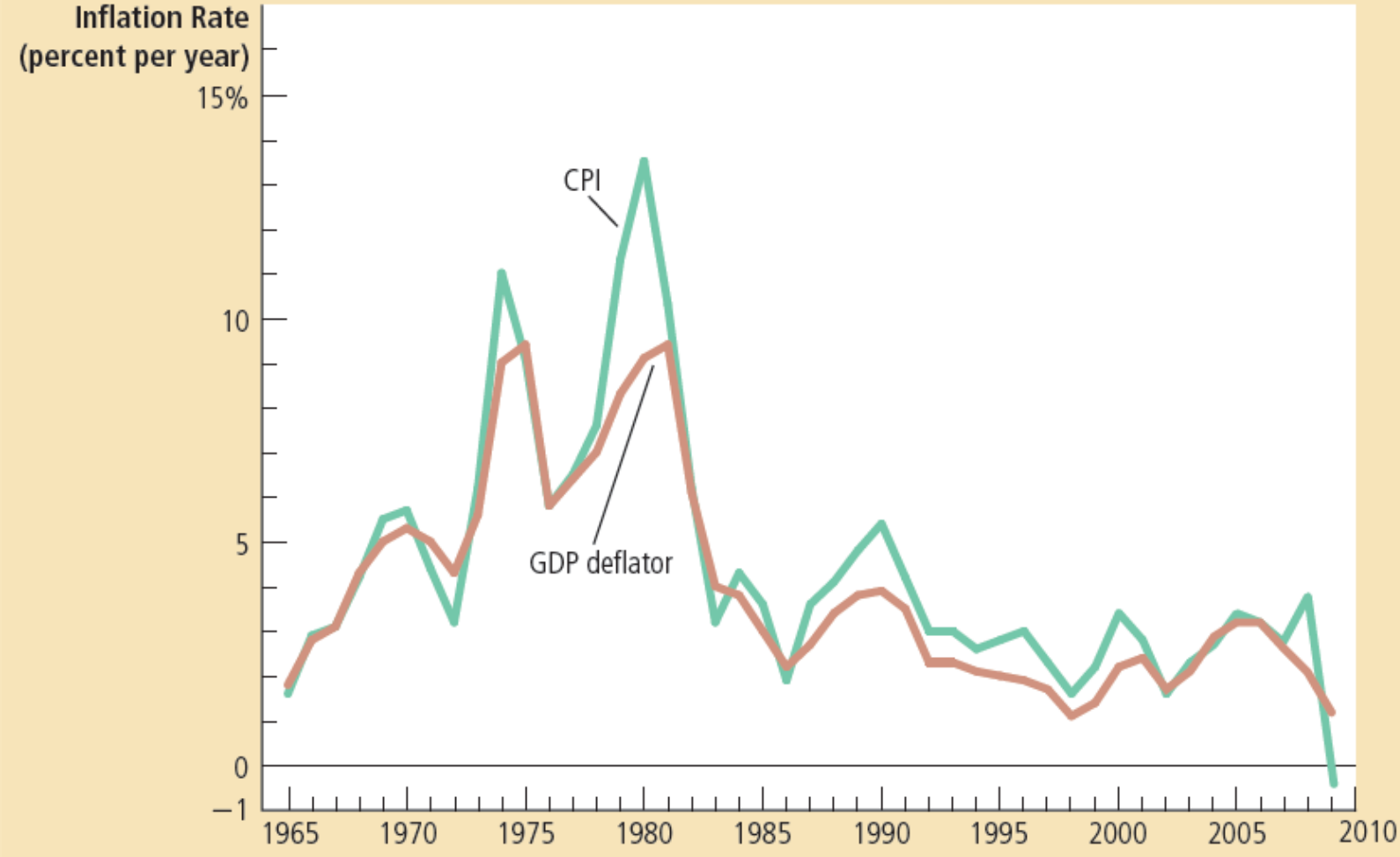
GDP Deflator vs CPI

- GDP Def considers entire basket of goods and services produced in the economy, CPI considers select goods, some of which might be imported (like oil)
- CPI basket is fixed over time, GDP Def basket changes automatically as production changes
  - according to each years demand/supply

Figure 2

Two Measures of Inflation



This figure shows the inflation rate—the percentage change in the level of prices— as measured by the GDP deflator and the consumer price index using annual data since 1965. Notice that the two measures of inflation generally move together.

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Correcting Economic Variables

- Dollar figures from different times

$$\text{Amount in today's dollars} =$$

$$= \text{Amount in year T dollars} \times \frac{\text{Price level today}}{\text{Price level in year T}}$$

- Indexation
  - Automatic correction by law or contract of a dollar amount for the effects of inflation
  - Social Security benefits

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Real and Nominal Interest Rates

- *Inflation is basically a fall in value of money.*
- Nominal interest rate
  - Interest rate as usually reported
  - Without a correction for the effects of inflation
- Real interest rate
  - Interest rate corrected for the effects of inflation

= Nominal interest rate – Inflation rate

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Interest rates in the U.S. Economy

- Nominal interest rate
  - Almost always exceeds the real interest rate
  - U.S. economy has experienced rising consumer prices in every year
  - Real interest rate exceeds the nominal interest rate in periods of deflation

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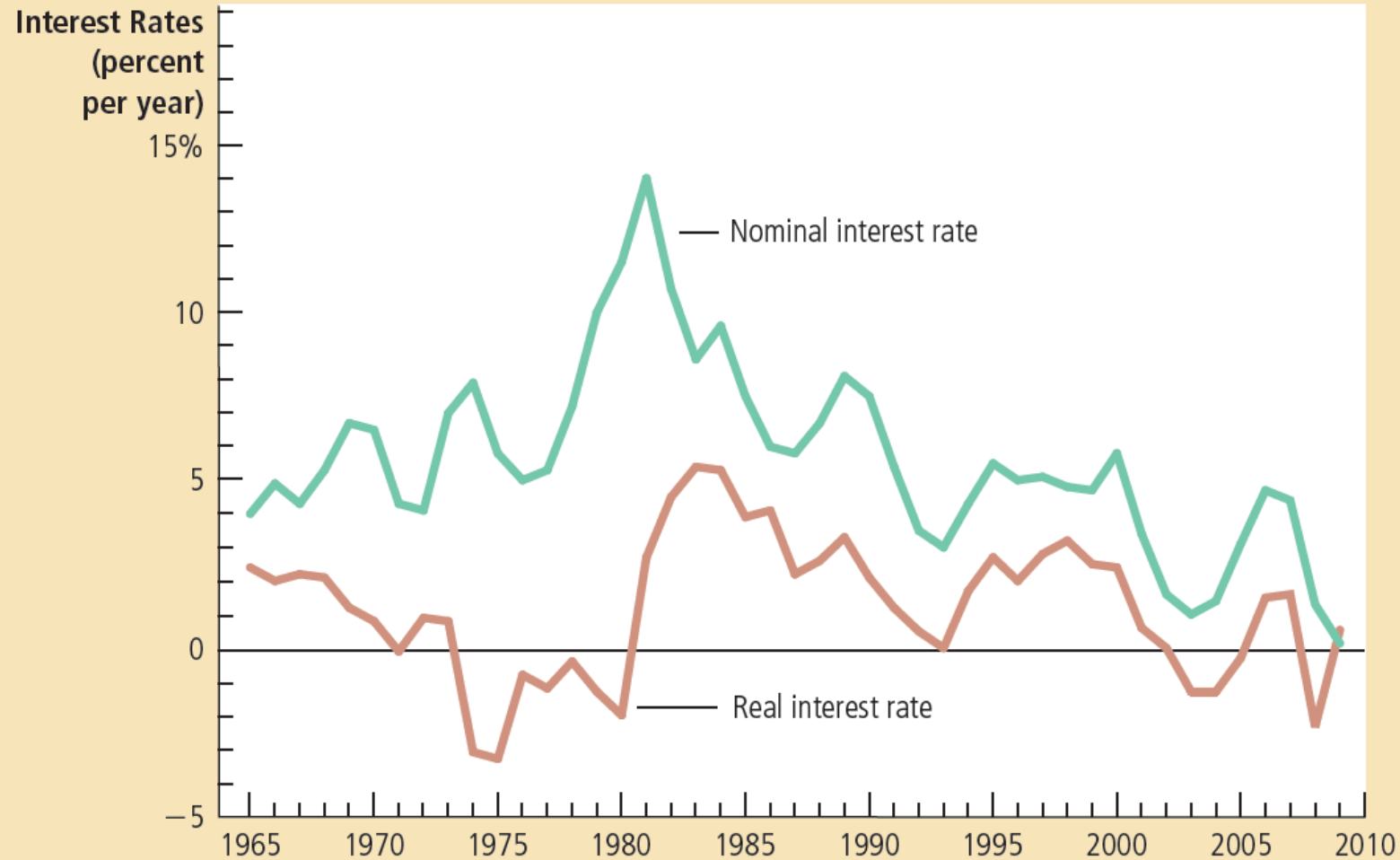
Figure 3

Real and Nominal Interest Rates



This figure shows nominal and real interest rates using annual data since 1965. The nominal interest rate is the rate on a 3-month Treasury bill. The real interest rate is the nominal interest rate minus the inflation rate as measured by the consumer price index. Notice that nominal and real interest rates often do not move together.

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